菲涅尔近似

(4.2.42)~(4.2.44)：

$$U(x_i,y_i)=\iint_{\infty} U(x_0,y_0)\,h(x_i,y_i;x_0,y_0)\,dx_0dy_0 \tag{4.2.42}$$

$$h(x_i,y_i;x_0,y_0)=h(P_i,P_0)=\frac{1}{i\lambda}\frac{e^{ikr}}{r}=\frac{1}{\lambda}\frac{e^{ikr-\frac{\pi}{2}}}{r} \tag{4.2.43}$$

$$r=\sqrt{(x_i-x_0)^2+(y_i-y_0)^2+z^2} \tag{4.2.44}$$

对(4.2.43)中 r 作二项式展开：

$$r=z\left\{1+\frac{(x_i-x_0)^2+(y_i-y_0)^2}{2z^2}-\frac{[(x_i-x_0)^2+(y_i-y_0)^2]^2}{8z^4}+\cdots\right\}$$

$$\approx z+\frac{(x_i-x_0)^2+(y_i-y_0)^2}{2z} \tag{4.4.2}$$

上式即 菲涅尔近似.

1. 菲涅尔的卷积积分

$$h(x_i-x_0,y_i-y_0)=\frac{e^{ikz}}{iz\lambda}e^{\frac{ik}{2z}[(x_i-x_0)^2+(y_i-y_0)^2]}$$

菲涅尔近似的物理实质为：用二次曲面代替球面的惠更斯子波.

$$U_i(x_i,y_i)=U_0(x_0,y_0)*h(x_i,y_i) \tag{4.4.4}$$

$$=\frac{e^{ikz}}{iz\lambda}\iint_{-\infty}^{\infty}U_0(x_0,y_0)\,e^{\frac{ik}{2z}[(x_i-x_0)^2+(y_i-y_0)^2]}dx_0dy_0$$

$$=\frac{e^{ikz}e^{\frac{ik}{2z}(x_i^2+y_i^2)}}{iz\lambda}\iint_{-\infty}^{\infty}U_0(x_0,y_0)\,e^{\frac{ik}{2z}(x_0^2+y_0^2)}e^{-\frac{i2\pi}{\lambda z}(x_0y_i+x_iy_0)}dx_0dy_0$$

令 $\xi=\frac{x_i}{\lambda z}$，$\eta=\frac{y_i}{\lambda z}$.

$$U_i(x_i,y_i)=\frac{e^{ikz}e^{\frac{ik}{2z}(x_i^2+y_i^2)}}{i\lambda z}\iint_{-\infty}^{\infty}U_0(x_0,y_0)\,e^{\frac{ik}{2z}(x_0^2+y_0^2)}e^{-2\pi(x_0\xi+\eta y_0)}dx_0dy_0$$

$$=\frac{e^{ikz}e^{\frac{ik}{2z}(x_i^2+y_i^2)}}{i\lambda z}F\left\{U_0(x_0,y_0)\,e^{\frac{ik}{2z}(x_0^2+y_0^2)}\right\}_{\xi=\frac{x_i}{\lambda z},\ \eta=\frac{y_i}{\lambda z}}\quad(4.4.7)$$

—— 菲涅尔衍射公式的 Fourier transform 形式.

2. 菲涅尔衍射近似程度.

菲涅尔近似

$$r\approx z+\frac{[(x_i-x_0)^2+(y_i-y_0)^2]}{2z}$$

是忽略了高阶项的. 这种近似所要求的精度大小对孔径，观察区域，以及距离 z 的线度有所限制.

当 $[(x_i-x_0)^2+(y_i-y_0)^2]$ 取最大时，$\Delta\varphi<1\,rad$.

$$\Delta\varphi=\frac{k[(x_i-x_0)^2+(y_i-y_0)^2]^2}{8z^3}\Big|_{max}\ll 1\,rad.$$

$$\Leftrightarrow\quad\frac{2\pi}{\lambda}\frac{[(x_i-x_0)^2+(y_i-y_0)^2]}{8z^3}\Big|_{max}\ll 1\,rad.$$

$$\Rightarrow\quad z^3\gg\frac{\pi}{4\lambda}[(x_i-x_0)^2+(y_i-y_0)^2]^2\Big|_{max}.$$

假定孔径最大径向范围为 D_0. $\sqrt{x_0^2+y_0^2}>D_0$，$U_0(x_0,y_0)=0$. 限定观察平面内的观测限制在一最大径向范围 D_i 内.

$$[(x_i-x_0)^2+(y_i-y_0)^2]\Big|_{max}=(D_0+D_i)^2.$$

$$\Rightarrow$$

$$z^3\gg\frac{\pi(D_0+D_i)^2}{4\lambda}.$$

3. 菲涅尔近似 $\Longleftrightarrow$ 傍轴近似.

菲涅尔近似下，(4.3.32) 中指数部份：

$$ikz\sqrt{1-(\lambda\xi)^2-(\lambda\eta)^2} \approx i\frac{2\pi}{\lambda}z\left[1-\frac{(\lambda\xi)^2+(\lambda\eta)^2}{2}\right] = ikz - i\pi\lambda z(\xi^2+\eta^2).$$

传递函数：

$$H(\xi,\eta) = \exp\left[ikz\sqrt{1-(\xi\lambda)^2-(\lambda\eta)^2}\right] = e^{ikz}e^{-i\pi\lambda z(\xi^2+\eta^2)}$$

相位延迟　　二次相位色散放在.

将 (4.3.32)，采用傍轴近似：$|\lambda\xi|<<1$，$|\lambda\eta|<<1$. 所得与上面完全一样.

∴ ~.

夫琅和费近似

若采用比菲涅尔近似更严格限制条件，z进一步增大，使：

$$\frac{k}{2z}(x_0^2+y_0^2)\Big|_{max} \ll 1 \text{ rad}.$$

$$\Leftrightarrow \qquad z \gg \frac{k}{2}(x_0^2+y_0^2)\Big|_{max} = \frac{\pi P_0^2}{\lambda}.$$

满足上面被称为夫琅和费衍射. 此z值区——夫琅和费衍射区.

如此，(4.4.7)

$$U_i(x_i,y_i) = \frac{e^{ikz}e^{\frac{ik}{2z}(x_i^2+y_i^2)}}{i\lambda z} F\left\{U_0(x_0,y_0)e^{\frac{ik}{2z}(x_0^2+y_0^2)}\right\}_{\xi=\frac{x_i}{\lambda z},\ \eta=\frac{y_i}{\lambda z}}.$$

中 $x_0^2+y_0^2$ 对相位影响可忽略.

$$U_i(x_i,y_i) = \frac{e^{ikz}}{i\lambda z} e^{\frac{ik}{2z}(x_i^2+y_i^2)} F\left\{U_0(x_0,y_0)\right\}_{\xi=\frac{x_i}{\lambda z},\ \eta=\frac{y_i}{\lambda z}}$$

$$= \frac{e^{ikz}}{i\lambda z} e^{i\pi\lambda z(x_i^2+y_i^2)} F\left\{U_0(x_0,y_0)\right\}_{\xi=\frac{x_i}{\lambda z},\ \eta=\frac{y_i}{\lambda z}}$$

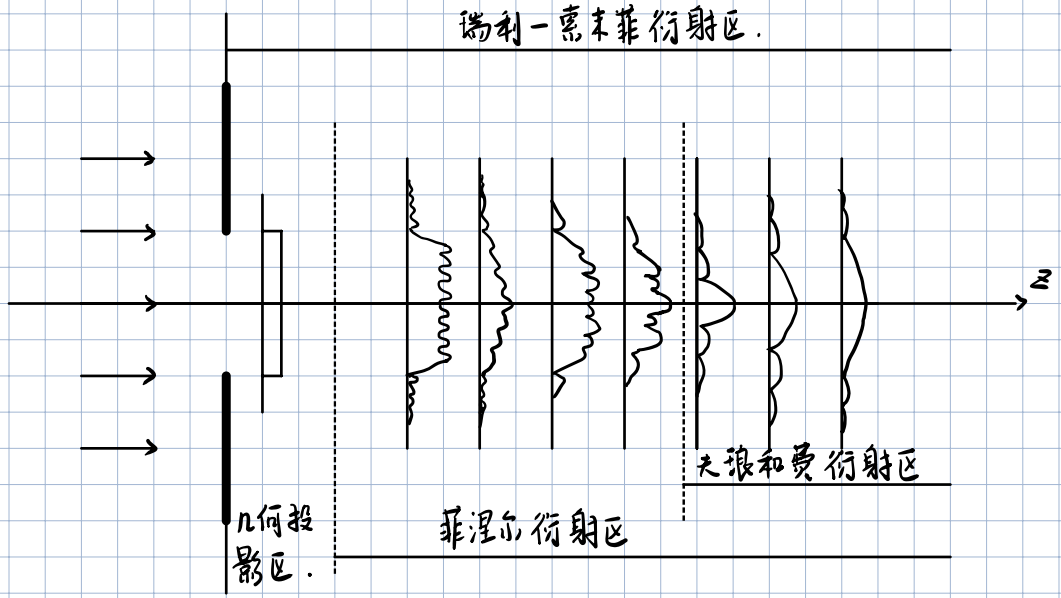
衍射光场光强分布：

$$I_i = |U_i(x_i,y_i)|^2 \propto \left(F\left\{U_0(x_0,y_0)\right\}_{\xi=\frac{x_i}{\lambda z},\ \eta=\frac{y_i}{\lambda z}}\right)^2.$$

夫琅和费衍射图样分布在系统的傅利叶变换平面上.

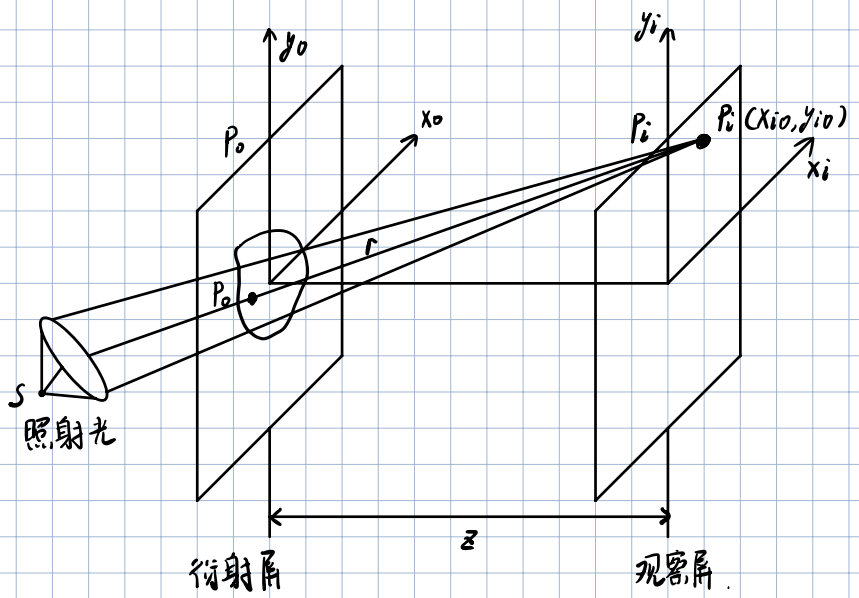
夫琅和费衍射条件实际上很苛刻。实验上可采用如会聚透镜消除变换函数中位相因子影响以实现近距离观察.

衍射区域划分



衍射屏被会聚球面波照射时的菲涅尔衍射

某些衍射问题中，如当照明衍射屏的为会聚球面波时，$U_0(x_0,y_0)$中将包含关于x_0,y_0的二次相位因子，在一定条件下可与(4.4.7)中$e^{\frac{ik}{2z}(x_0^2+y_0^2)}$相消.



$$r=\mathrm{sgn}z\left[z^2+(x_0-x_{i0})^2+(y_0-y_{i0})^2\right]\approx z+\frac{(x_0-x_{i0})^2}{2z}+\frac{(y_0-y_{i0})^2}{2z}$$

若无衍射屏，由P_i发出球面波自由传播至P_0处光场复振幅为：

$$U_0(x_0,y_0)=\frac{A_0}{z}e^{-kz}e^{-\frac{ik}{2z}\left[(x_0-x_{i0})^2+(y_0-y_{i0})^2\right]}$$

设屏函数$t(x_0,y_0)$.

$$U_0'(x_0,y_0)=t(x_0,y_0)\frac{A_0}{z}e^{-kz}e^{-\frac{ik}{2z}\left[(x_0-x_{i0})^2+(y_0-y_{i0})^2\right]}$$

如此，观察屏上任一点复振幅：

$$U_i(x_i,y_i)=\frac{e^{ikz}e^{\frac{ik}{2z}(x_i^2+y_i^2)}}{i2\lambda}\iint_{-\infty}^{+\infty}U_0'(x_0,y_0)e^{\frac{ik}{2z}(x_0^2+y_0^2)}e^{-\frac{i2\pi}{\lambda z}(x_0x_i+y_0y_i)}dx_0dy_0$$

$$= \frac{e^{ikz}e^{\frac{ik}{2z}(x_i^2+y_i^2)}}{iz\lambda}\iint_{-\infty}^{\infty} t(x_0,y_0)\frac{A_0}{z}e^{-ikz}e^{-\frac{ik}{2z}[(x_0-x_{i0})^2+(y_0-y_{i0})^2]}e^{\frac{ik}{2z}(x_0^2+y_0^2)^2}e^{-\frac{i2\pi}{\lambda z}(x_0x_i+y_0y_i)}dx_0dy_0$$

$$= \frac{A_0}{i\lambda z^2}e^{-\frac{ik}{2z}(x_{i0}^2+y_{i0}^2)}e^{\frac{ik}{2z}(x_i^2+y_i^2)}\iint_{-\infty}^{\infty} t(x_0,y_0)e^{-i2\pi(\frac{x_i-x_{i0}}{\lambda z}x_0+\frac{y_i-y_{i0}}{\lambda z}y_0)}dx_0dy_0$$

$$= \frac{A_0}{i\lambda z^2}e^{-\frac{ik}{2z}(x_{i0}^2+y_{i0}^2)}e^{\frac{ik}{2z}(x_i^2+y_i^2)}T(\xi,\eta).$$

其中，$\xi=\frac{x_i-x_{i0}}{\lambda z}$，$\eta=\frac{y_i-y_{i0}}{\lambda z}$　$T(\xi,\eta)=F\{t(x_0,y_0)\}$.

光场强度：

$$I_i(x_i,y_i)=|U_i(x_i,y_i)|^2=(\frac{A_0}{\lambda z^2})^2T^2(\xi,\eta).$$

与夫琅和费衍射相同.

衍射的巴俾涅原理

互补屏：Σ_1 开孔部份与 Σ_2 不透明部份对应.

$$t_2(x_0,y_0) = 1 - t_1(x_0,y_0).$$

$$\Sigma_1 + \Sigma_2 = \Sigma_\infty = \infty$$

菲涅尔-基尔霍夫公式：

$$\frac{1}{i\lambda}\iint_{\Sigma_1} U_0(P_0)K(\theta)\frac{e^{ikr}}{r}ds_0 + \frac{1}{i\lambda}\iint_{\Sigma_2} U_0(P_0)K(\theta)\frac{e^{ikr}}{r}ds_0 = \frac{1}{i\lambda}\iint_{\infty} U_0(P_0)K(\theta)\frac{e^{ikr}}{r}ds_0$$

即：

$$U_{i1}(P_i) + U_{i2}(P_i) = U_i(P_i).$$

$U_{i1}(P_i)$ 与 $U_{i2}(P_i)$ 分别表衍射屏 Σ_1 与 Σ_2 开孔区域在观察平面 P_i 上的光场，而 2 屏开孔部分加起来不存在不透明区域.

∴ 两屏在 P_i 产生衍射光场，其振幅之和等于光波自由传播时该点复振幅之和.

——巴俾涅原理

互补屏 ⟶ 孔径无限大衍射屏.